

California Housing

Price Prediction

Linear Regression Project | ML Lifecycle Demonstration

Python

pandas

scikit-learn

matplotlib

seaborn

~50.7K

MAE

~70K

RMSE

~0.63

R^2

Project Overview

OBJECTIVE

Predict California housing prices using a Linear Regression model trained on the classic California Housing dataset.

DATASET

California Housing dataset (Kaggle CSV). 20,640 rows with housing-related features including income, rooms, population, and location.

TARGET VARIABLE

median_house_value

Skills Demonstrated



EDA

Distributions, correlations, missing value analysis



Preprocessing

One-hot encoding, mean imputation, feature selection



Model Training

Linear Regression with 80/20 train-test split



Evaluation

MAE, RMSE, R^2 score metrics



Visualization

Scatter plots, heatmap, residual distribution



Reporting

Jupyter Notebook + PDF slide deck

Step-by-Step Workflow

01 Environment Setup

Installed pandas, numpy, scikit-learn, matplotlib, seaborn, jupyter

02 Load Dataset

Used `pd.read_csv()` with California Housing CSV; verified shape (20,640 × 10)

03 EDA

Performed exploratory data analysis using `df.head()`, `df.info()`, `df.describe()`, and correlation heatmap

04 Preprocessing

Applied one-hot encoding using `pd.get_dummies()` and handled missing values using mean imputation

05 Train/Test Split

80% train / 20% test using
`train_test_split(random_state=42)`

06 Train Model

`LinearRegression().fit(X_train, y_train)`

07 Evaluate

Computed MAE, RMSE, R^2 on held-out test set

EDA & Preprocessing

EXPLORATORY DATA ANALYSIS

20,640 total records in dataset

13 input features after encoding

207 missing values in
total_bedrooms

0.69 strongest correlation:
median_income vs
median_house_value

\$206K median target value

PREPROCESSING PIPELINE

Handle Categoricals

```
pd.get_dummies(df, drop_first=True)
```

Converts ocean_proximity into numeric dummy columns

Fill Missing Values

```
df.fillna(df.mean())
```

Imputes 207 NaN rows in total_bedrooms

Correlation Heatmap

```
sns.heatmap(df.corr(), cmap='coolwarm')
```

Identifies feature relationships

Feature/Target Split

```
X = df.drop('median_house_value')  
y = df['median_house_value']
```

Model Evaluation Results

MAE

~\$50,700

Mean Absolute Error

Average dollar deviation per prediction

RMSE

~\$70,000

Root Mean Squared Error

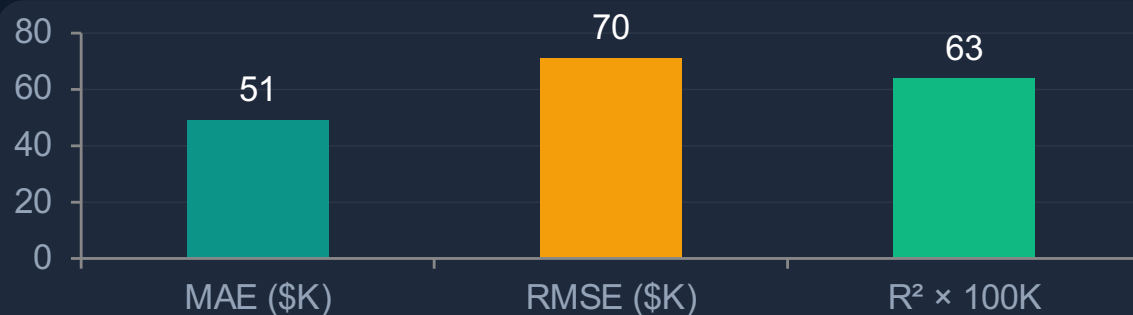
Penalizes large errors more heavily

R²

~0.63

R² Score

63% of price variance explained by the model



Interpretation

R² of 0.63 is a solid baseline for Linear Regression on this dataset. The model captures the main pricing trends but leaves room for improvement with tree-based models.

Visualizations Generated



Correlation Heatmap

```
sns.heatmap(df.corr(), cmap='coolwarm')
```

Insight

Reveals that median_income has the strongest positive correlation with house value ($r \approx 0.69$). Latitude shows moderate negative correlation with house prices.



Actual vs Predicted

```
plt.scatter(y_test, y_pred)
```

Insight

Points closer to the diagonal line indicate better prediction accuracy. Spread at higher values shows the model underestimates expensive homes.



Residual Distribution

```
sns.histplot(residuals, bins=50, kde=True)
```

Insight

Residuals are centered near 0, indicating reasonably balanced prediction errors. Slight right skew from high-value homes.

Key Learnings & Conclusion



Full ML Lifecycle

Successfully executed end-to-end: data loading → EDA → preprocessing → training → evaluation → reporting



Model Limitations

Linear Regression achieves $R^2 \approx 0.63$. Housing prices have non-linear patterns better suited for Random Forest or XGBoost



Data Quality Matters

207 missing values in total_bedrooms required mean imputation; ocean_proximity needed one-hot encoding



Feature Importance

median_income emerged as the strongest predictor (correlation ≈ 0.69), validating economic intuition



Next Steps

Feature engineering, polynomial features, regularization (Ridge/Lasso), or ensemble methods to improve R^2